Characterizing Resonance Behavior in Electronic and Mechanical Oscillators

Julian Woo*

PHY324 University of Toronto

Abstract

In this experiment we tested resonance in two oscillatory systems: an electronic series LCR circuit and a mechanical tuning fork, using transient ring-downs and frequency sweeps of amplitude ratio and phase to extract the resonant frequency f_0 and quality factor Q. For the LCR circuit, all three methods gave consistent results with $f_0 \approx 4.9$–5.0 kHz and $Q \approx 2.5$–2.7, slightly lower than the theoretical prediction $Q_{theory} \approx 3.0$, indicating additional unmodelled series resistance. For the tuning fork, the ring-down analysis gave $f_0 = (75.379 \pm 0.006)$ Hz and $Q = 137.7 \pm 0.4$, while driven amplitude and phase measurements, taken on a different day with the setup rebuilt, yielded consistent values of $f_0 \approx 43$ kHz and $Q \approx 18$ that do not match the ring-down result, likely because the driven circuit and coil configuration were not reconstructed correctly. Overall, we determined f_0 and Q for both systems and showed that, despite their physical differences, the electrical and mechanical oscillators are described by the same damped–driven oscillator model, with parasitic losses and setup choices setting the accuracy of our measured parameters.

1 Introduction

Oscillatory motion is studied by looking at both a mechanical oscillator (a tuning fork) and an electrical circuit oscillator (an LCR circuit). The equation that describes all oscillations with damping and no driving force is:

$$x(t) = Ae^{-\gamma t}\cos(\omega_d t + \phi) \tag{1}$$

(Lab Manual - 6, with $\gamma = \frac{1}{\tau}$), where $\gamma = b/(2m)$ is the damping coefficient and $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ is the damped angular frequency with ω_0 being the resonant frequency. The exponential term is the envelope of the oscillation, and represents the rate at which energy dissipates from the system.

1.1 Driven Oscillations and Resonance

To sustain oscillation despite damping, an external driving force is applied of the form:

$$F(t) = F_0\cos(\omega t) \tag{2}$$

with driving amplitude F_0 and driving angular frequency ω. Incorporating this into the equation we get:

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = F_0\cos(\omega t). \tag{3}$$

The solution describes an oscillation at the same frequency as the driving force but with a different amplitude A and phase shift ϕ:

$$x(t) = A(\omega)\cos(\omega t + \phi). \tag{4}$$

The amplitude as a function of frequency is given by

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}. \tag{5}$$

Equation (5) shows that the amplitude reaches a maximum when $\omega \approx \omega_0$, a condition known as *resonance*. Resonance occurs when the driving frequency matches the system's natural frequency, allowing maximum energy transfer from the driver to the oscillator.

1.2 The Quality Factor Q

The *quality factor*, or Q, quantifies how "sharp" or "selective" this resonance is and provides a measure of energy retention versus loss per cycle. A system with a higher Q oscillates longer before its energy dissipates. The quality factor is defined as:

$$Q = 2\pi \times \frac{\text{Energy stored}}{\text{Energy lost per cycle}}. \tag{6}$$

For a lightly damped oscillator, this can be expressed in terms of parameters as:

$$Q = \frac{m\omega_0}{b} = \frac{\omega_0}{2\gamma}. \tag{7}$$

Alternatively, when resonance is measured experimentally from amplitude–frequency data, the Q factor is determined by the ratio of the resonant frequency ω_0 to the full width at half maximum (FWHM) $\Delta\omega$ of the resonance peak:

$$Q = \frac{\omega_0}{\Delta\omega}. \tag{8}$$

This allows a comparison between theoretical and experimental results for both electrical and mechanical systems.

1.3 Electrical and Mechanical Analogies

The same mathematics that describe mechanical oscillators also apply to electrical systems composed

*julian.woo@mail.utoronto.ca

of inductors, capacitors, and resistors—known as *LCR circuits*. In an LCR circuit, the charge q on the capacitor obeys an analogous differential equation:

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = V_0\cos(\omega t) \tag{9}$$

where L is the inductance, R the resistance, C the capacitance, and V_0 the driving voltage amplitude. Here, the mechanical quantities m, b, and k correspond to L, R, and $1/C$ respectively. Thus, the current $I = dq/dt$ behaves analogously to the velocity of the mass, and the voltage corresponds to force.

Just as with a mechanical oscillator, the LCR circuit exhibits a resonance peak at

$$\omega_0 = \frac{1}{\sqrt{LC}}, \tag{10}$$

and its quality factor is

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}. \tag{11}$$

The electrical setup therefore provides a convenient and precise way to measure resonance and damping electronically before exploring analogous effects in a mechanical system such as a *tuning fork oscillator*, which converts magnetic and mechanical energy in similar fashion.

1.4 Purpose and Context

The study of oscillators and their quality factors is central to understanding how systems store and dissipate energy—concepts that extend from radio circuits to seismology and quantum physics. In this experiment, we investigate the behavior of both an electrical LCR circuit and a mechanical tuning fork oscillator to determine their respective quality factors and to explore how resistance and friction influence energy loss and resonance width. By comparing the results from two distinct physical systems governed by equivalent mathematical frameworks, we gain insight into the universal nature of resonance and damping.

2 Methods and Procedures

2.1 Exercise 1: Electronic Oscillator

In this experiment, using an Oscilloscope (Keysight DSOX-1202G), 10 nF Capacitor, 1 kΩ resistor, and a 114 mH inductor we built the circuit shown in Figure 1. Then we measured the Capacitance, inductance, and resistance of each component, as seen in the table below: We noted down the oscillations on the measurement devices for later uncertaintyp calculations. We then drove the LCR circuit with a square wave and measured the oscillation frequency after each transition of the square wave.

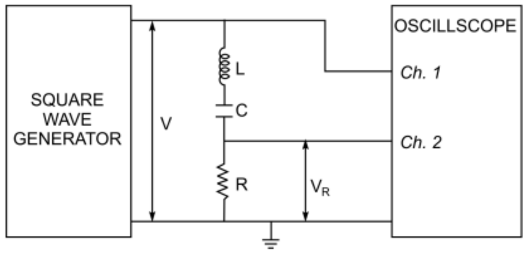

Figure 1: *Experimental Setup for Exercise 1. circuit with a wave generator (far left) Inductor (L), Capacitor (C) and resistor (R) in series, and an oscilloscope in parallel (far right)*

Capacitor (C)	$9.9 \cdot 10^{-9} \pm 3 \cdot 10^{-10} F$
Inductor (L)	0.114 ± 0.009 H
Resistor (R)	$1073\Omega \pm 6\ \Omega$

Table 1: *Measured Component Values*

Then we used the signal generator to generate sine waves, and starting at 400 Hz, ending at 6000Hz and going up in 400Hz increments, we measure the actual frequency (not the setting), Input voltage (voltage around the whole circuit) and its uncertainty, the output voltage (voltage across only the resistor) and its uncertainty, and the phase and its uncertainty using the oscilloscopes measure settings for phase, and amplitude. We repeated these measurements for each Frequency value.

After this and logging between what two values of frequency the highest output voltage lies, we then used a value right below the lower bound as the starting point (4750Hz) and the upper bound as the ending (5000Hz), and went up in 25Hz increments to find the resonant frequency more accurately, and we noted down the highest value of output voltage. We also repeated the earlier measurements for each Frequency value.

2.2 Exercise 2: Mechanical Oscillator

The capacitor and resistor were removed and only the inductor was kept in the circuit. We then set the oscilloscope to 2 seconds per division and hit the tuning fork with a solid object with high force. When the screen updated and there was enough time periods visible to show the decay clearly, we clicked the stop/run button to pause the measurements and exported the data to a usb.

We then connected the signal generator to the driving coil that is between the tuning fork ends, and repeated the same steps as for exercise 1, measuring the input voltage from the pickup coil for each frequency, the phase and the output voltage of the signal generator. We started at 10kHz, ended at 55kHz and went up in 2.5kHz increments, and then to find the resonant frequency more accurately, we then used a value right below the lower bound as the starting point (42.5kHz) and the upper bound as the ending (45kHz), and went up in 0.25kHz.

3 Analysis and Discussion

3.1 Electrical Oscillator Analysis

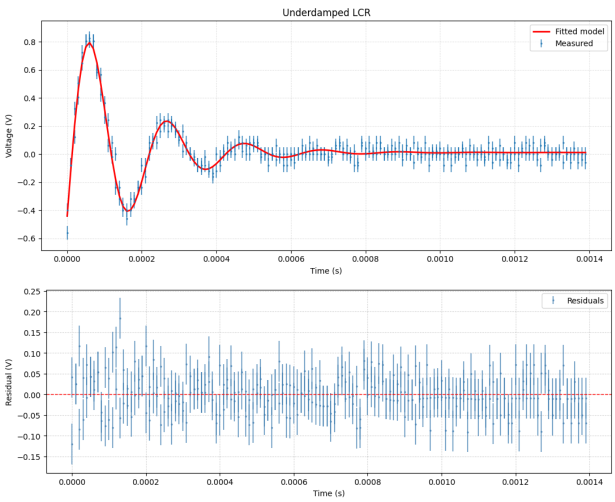

Figure 2: ***Top)*** *Measured voltage response of the underdamped LCR circuit following a square-wave transition, fitted with the underdamped model. The best-fit parameters are* $\tau = (0.17 \pm 0.01)$ ms*,* $f_0 = (4810 \pm 60)$ Hz*, and* $Q = 2.5 \pm 0.2$*. The fit achieves a reduced chi-squared value of* $\chi^2 = 0.87$ *with 275 degrees of freedom,* ***Bottom)*** *Residuals of the fit showing random scatter about zero.*

For *Exercise 1*, we first drove the series RLC with a square wave to verify that we have an underdamped wave, then used the oscilloscope's wave generator to measure the amplitude ratio $H_a(f) = V_R/V_{\text{in}}$ and the phase $\theta(f)$ across the resistor.

For the transient check, we drove the series RLC with a square wave to confirm it was underdamped and to obtain a estimate of f_0 and Q. We then Fit the underdamped model to obtain τ and f_0, For the underdamped oscillator, the response is well described by

$$V(t) = A\,e^{-t/\tau}\cos(\omega t + \phi) + C,$$

Fitting the waveform in Fig. 2 (top) gives

$$\tau = (0.17 \pm 0.01)\text{ ms}$$
$$f_0 = (4810 \pm 60)\text{ Hz}$$
$$Q = 2.5 \pm 0.2$$

with $\chi^2_{\text{red}} = 0.87$ for 275 dof. Residuals (Fig.2) are random about zero, indicating the model is adequate. The circuit is clearly underdamped; $f_{0,\text{ring}} \approx 4.81$ kHz is consistent with the later sweep; and Q value near 2.5 sets expectations for what we might expect later

3.2 Amplitude Ratio vs Frequency

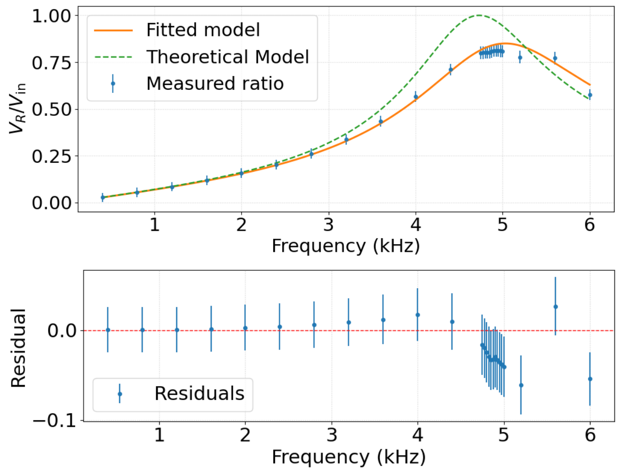

Figure 3: ***Top)*** *Measured amplitude ratio* V_R/V_{in} *versus frequency for the series RLC circuit driven by a sine source, fitted with the magnitude model The theoretical response computed from independently measured* L, R, C *is shown as a green dashed curve. The best–fit parameters are* $f_0 = (5030 \pm 40)$ *Hz,* $Q = 2.6 \pm 0.1$*, and an inter-channel scale factor* $A = 0.85 \pm 0.01$*. The fit yields* $\chi^2 = 17.84$ *for 22 degrees of freedom, giving a reduced chi-squared of* $\chi^2_{red} = 0.81$*.* ***Bottom)*** *Residuals of the fit*

We swept the drive frequency with the scope's wave generator and measured $H(f) = V_R/V_{\text{in}}$ across the series resistor. The resistor was chosen so the standard series–RLC transfer function applies directly. We collected (f, V_{in}, V_R) from 0.4–6 kHz, propagated voltage and frequency uncertainties, and plotted $H(f)$ with error bars (Fig.3, top) and residuals (Fig.3, bottom). We performed a nonlinear fit on the resistor-branch magnitude

$$H(\omega) = A\Big[1 + \big(Q(\omega/\omega_0 - \omega_0/\omega)\big)^2\Big]^{-1/2}$$

report $f_0, Q, A,\ \chi^2/\chi^2_{\text{red}}$, and interpret residuals. Compare the fit to the ideal response computed from metered L, C, R. For a driven series RLC, andwidth encodes Q; fitting the full curve uses all points with their uncertainties and is more precise than using only the half-power. The fit in Fig.3 yields

$$f_{0,\text{amp}} = (5030 \pm 40)\text{ Hz} \tag{12}$$
$$Q_{\text{fit}} = 2.6 \pm 0.1 \tag{13}$$
$$A = 0.85 \pm 0.01, \tag{14}$$

with $\chi^2 = 17.84$ for 22 dof ($\chi^2_{\text{red}} = 0.81$). The ideal metered-parts curve (green dashed) peaks higher/narrower than the data, whereas the fit follows the measured, slightly broadened peak. Residuals are small in the wings and show a negative dip centered near resonance, indicating modest systematic broadening relative to the ideal model. The amplitude data has a Q near 2.6, consistent with the transient estimate and noticeably lower than the prediction from the metered components, signaling potential extra series loss in the real setup.

3.3 Phase vs Frequency

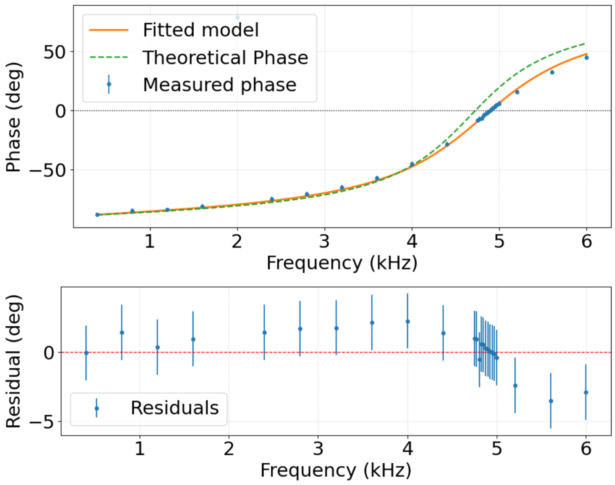

Figure 4: ***Top)*** *Measured phase response of the series LCR circuit versus frequency, fit with the velocity-phase model, and compared to the theoretical phase from measured L, R, C. The best-fit parameters are $f_0 = (4901.8$ Hz) and $Q = 2.69$. The fit yields $\chi^2 = 13.18$ for 22 degrees of freedom, giving a reduced chi-squared of $\chi^2_{red} = 0.60$.* ***Bottom)*** *Residuals of the fit with $\pm\sigma$ error bars*

Using V_{in} as the reference channel, we measured the phase of V_R versus frequency and analyzed it with the same oscillator parameters used for the magnitude. We plotted $\theta(f)$ with error bars (Fig.4, top) and the corresponding residuals (Fig.4, bottom). A theoretical phase from metered L, C, R is plotted for context.

Fit the velocity phase model

$$\phi(f) = \arctan\Big(\big(Q(\omega/\omega_0 - \omega_0/\omega)\big)^{-1} \Big) \quad (\phi_0 = 0),$$

and report $f_0, Q, \chi^2/\chi^2_{\text{red}}$, and residuals; then check consistency with the amplitude-derived parameters. The same (ω_0, Q) that shapes the magnitude also governs the phase's arctangent; agreement between the two serves as an internal cross-check. The phase fit (Fig.4) returns

$$f_0 = 4902 \pm 9 \text{ Hz}, \qquad Q = 2.69 \pm 0.08,$$

Residuals are structureless within the quoted uncertainties and the fitted arctangent tracks the measured phase through the expected $+90^\circ \to 0^\circ \to -90^\circ$ transition near resonance. The phase-derived f_0 and Q agree with those obtained from the amplitude fit and transient response within 1σ–2σ, providing an internal cross-check that the same (ω_0, Q) shows both magnitude and phase. Phase independently supports a Q near 2.6–2.7 and an f_0 close to 4.9–5.0 kHz, consistent with the amplitude and transient analyses.

3.4 Analysis vs Theory

Using measured components,

$$Q_{\text{theory}} = \frac{1}{R}\sqrt{\frac{L}{C}} = 3.2 \pm 0.1, \tag{15}$$

$$f_{0,\text{theory}} = \frac{1}{2\pi\sqrt{LC}} = 4.7 \pm 0.2 \text{ kHz}. \tag{16}$$

Because the wave generator behaves as a 50 Ω source, our baseline uses $R_{\text{eff}} = R + 50 = 1123\ \Omega$, giving

$$Q_{\text{theory}} = Q_{\text{theory}} \frac{R}{R+50} \approx 3.02.$$

Our amplitude result from the amplitude ratio is $Q = 2.6 \pm 0.1$,which is lower by $\Delta Q = -0.46$ ($\sim 2.7\sigma$ with the same combined uncertainty). Interpreting $Q = (1/R_{\text{eff}})\sqrt{L/C}$, the data implies $R_{\text{eff}} \approx 1326\ \Omega$, $\Delta R_{\text{loss}} \approx 200\ \Omega$ beyond $R + 50\ \Omega$, which is fairly consistent with inductor loss and small lead resistance at ~ 5 kHz. The depressed peak in Fig.3 and its residual dip at resonance are good evidence of added series loss.

3.5 Mechanical Oscillator Analysis

In this experiment we drove a mechanical oscillator and characterized its response in the time and frequency domains. We first verified underdamped behavior from a ring–down, then plotted the amplitude ratio $R(f) = V_{\text{r}}/V_{\text{in}}$ and the phase $\phi(f)$ versus frequency both using measured data.

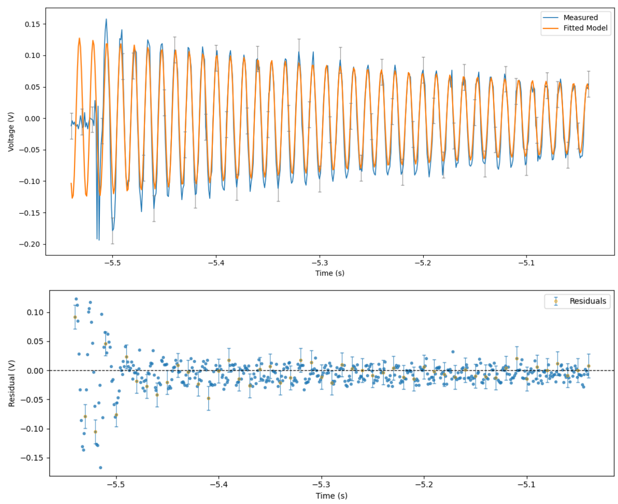

Figure 5: ***Top)*** *Ring-down of the tuning fork fit with underdamped model. The best-fit parameters are $f = (75.379 \pm 0.006$ Hz) ($\omega = 473.62 \pm 0.04$ rad/s), $\tau = 0.5814 \pm 0.0006$ s, and $Q = 137.7 \pm 0.4$. The fit yields $\chi^2 = 1229.68$ for 1995 degrees of freedom and a reduced chi-squared of $\chi^2_{red} = 0.616$* ***Bottom)*** *Residuals of the fit.*

We excited the fork and recorded voltage versus time data on a usb stick to estimate the dissipation time constant and natural frequency. The underdamped model

$$y(t) = A\, e^{-t/\tau} \cos(\omega t + \phi) + C$$

was fitted to the waveform in Fig. 5 (top).

The fit reports $\chi^2 = 1229.68$ for 1995 degrees of freedom, i.e. $\chi^2 = 0.616$. Residuals (Fig. 5, bottom) are random about zero, indicating that the underdamped model adequately describes the decay and providing a benchmark Q for the driven measurements. The mechanical system exhibits remarkably low damping, with a Q factor more than 50 times higher than the electronic RLC circuit from Exercise 1, consistent with our expectations for a steel tuning fork that the Q factor should be much larger.

3.6 Amplitude Ratio vs Frequency

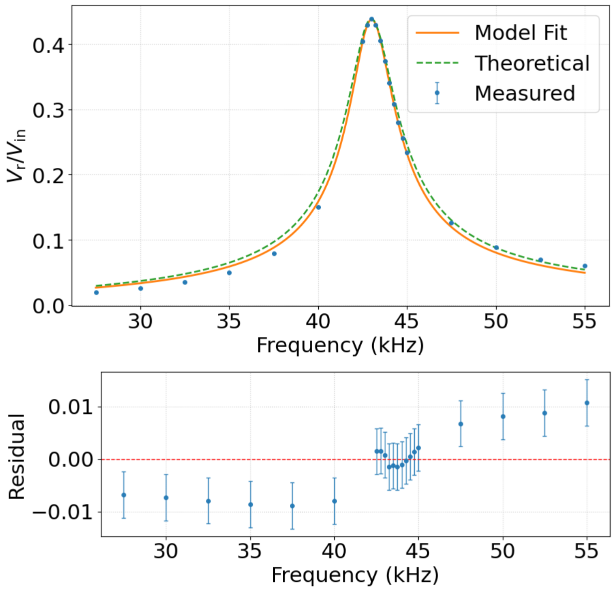

Figure 6: ***Top)*** *Measured amplitude ratio of the driven resonator, V_{out}/V_{in}, plotted versus frequency, yielding best–fit parameters $f_0 = (43.014 \pm 0.028)$ kHz, $Q = 17.73 \pm 0.56$, and $A = 0.4379 \pm 0.0032$. The fit achieves a reduced chi–squared of $\chi^2 = 2.03$ with 18 degrees of freedom.* ***Bottom)*** *Residuals of the fit versus frequency.*

We swept the drive frequency with the scope's generator and measured the amplitude ratio $R(f) = V_r/V_{in}$ using the AC RMS readout on the oscilloscope. For each point we logged (f, V_r, V_{out}) in the 26–55 kHz band and propagated uncertainties: the channel uncertainties were combined with a Keysight volts/div uncertainty. We fit the data with the resonance model

$$R(f) = \frac{A}{\sqrt{1 + Q^2\big((\omega/\omega_0) - (\omega_0/\omega)\big)^2}},$$

The residuals (Fig. 6, bottom) show no systematic pattern, indicating that this model captures the overall shape of the amplitude response. The reduced chi–squared value of $\chi^2 = 2.03$ suggests that the fit is reasonably good, though the scatter near the peak is somewhat larger than expected from our error bars, implying that we may slightly underestimate the uncertainties close to resonance. The Q factor from this measurement is also different from the ring-down value but remains considerably higher than typical electronic circuits, characteristic of mechanical resonators with low energy dissipation.

We used a theoretical model derived from our data to generate the theoretical curve For the theoretical amplitude ratio curve shown in Fig. 6, we used a half-power method based on the experimental data. Using peak detection to find the maximum amplitude, we fit a parabola to the top three points around the maximum and took the vertex as the resonant frequency. Moreover, we found where the amplitude falls to $R_{max}/\sqrt{2}$ by interpolating between the nearest points on each side of the peak, obtaining f_L and f_R. calculated $Q_{th} = \frac{f_{0,th}}{f_R - f_L}$ directly from the bandwidth. Using these parameters, we constructed the theoretical curve shown in figure 6

3.7 Phase vs Frequency

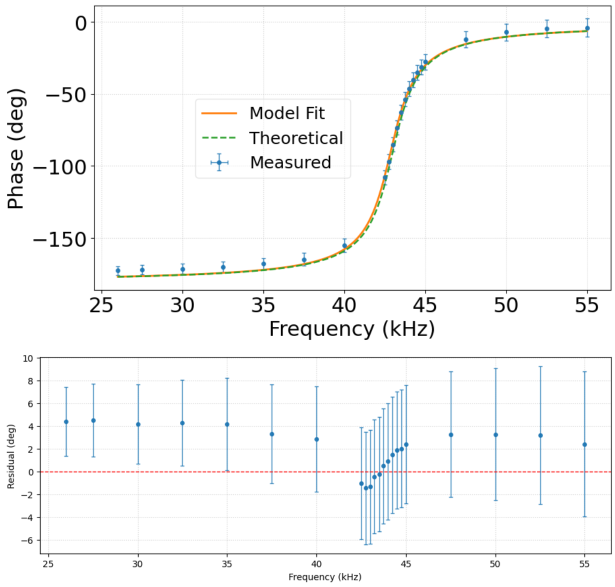

Figure 7: ***Top)*** *Measured phase response of the driven resonator, $\phi(f)$ in degrees, plotted versus drive frequency. The solid curve shows a fit to the velocity–phase model, yielding $f_0 = (42.867 \pm 0.050)$ kHz and $Q = 17.71 \pm 1.21$. The fit attains a chi–squared of $\chi^2 = 10.68$ with 20 degrees of freedom, corresponding to a reduced chi–squared of $\chi^2 = 0.53$.* ***Bottom)*** *Residuals of the fit plotted versus frequency.*

We also analyzed the phase relationship between the input drive and pickup coil output across the same frequency range. For a velocity-based detection like our pickup coil, we expected the phase response to follow

$$\phi(f) = -\arctan\Big(Q\big(\omega/\omega_0 - \omega_0/\omega\big)\Big), \qquad \omega = 2\pi f,$$

Fitting this model to the measured phase data (Fig. 7, top) yielded $f_0 = (42.867 \pm 0.050)$ kHz, $Q = 17.71 \pm$

$1.21, with \chi^2 = 0.53 for 20$ degrees of freedom. The reduced chi–squared significantly below one suggests that our phase uncertainties are somewhat overestimated, but the very small residuals (Fig. 7, bottom) and the clean S–shaped phase transition confirm that the arctangent model accurately describes the data.

Using parameters consistent with the amplitude analysis in the same phase formula produces the dashed theoretical curve in Fig. 7. The fact that this curve is visually indistinguishable from the best–fit phase model shows that the parameters extracted from amplitude and phase are mutually consistent. The phase results therefore reinforce our conclusion that the mechanical oscillator has $Q \approx 18$ in this configuration.

3.8 Uncertainty Analysis

The horizontal time base uncertainty is in nanoseconds so we ignore that in our uncertainties in the plots and analysis. A significant source of experimental uncertainty was inadvertent vibrations introduced by our movement in proximity to the tuning fork and LCR apparatus. Even minor movements in the laboratory environment could induce parasitic oscillations or damping effects in this mechanical resonator system. This likely contributed to the difficulties in obtaining phase and voltage data. In addition, mechanical coupling between the tuning fork and its mounting structure,non-ideal behavior of the pickup and drive coils, position dependent coupling between the fork and the coils. Finally, it is possible that the driven circuit was not implemented correcly which would have a significant effect on the amplitude and phase responses. This likely contributed to our sources of error and our discrepancies in values between our ring-down graph and our amplitude and phase response values. Despite these limitations, the amplitude–ratio data produced a clean resonance curve with reasonable fit statistics, and the phase data were well described by the velocity–phase model, suggesting that these systematic effects did not dominate our final results.

4 Conclusion

In this experiment we studied two oscillatory systems an electronic series LCR circuit and a mechanical tuning fork – using ring–down measurements and frequency–domain amplitude and phase sweeps to determine their resonant frequencies and quality factors and to test the damped, driven–oscillator model.

For the electronic LCR circuit, the transient ring–down, amplitude–ratio sweep, and phase–frequency sweep all returned consistent values of f_0 (around 5 kHz) and Q (around 2.6). Reduced chi–squared values were close to one and residuals were structureless, so the standard second–order transfer functions accurately described the data. When compared with the theoretical Q computed from the metered component values, our measured Q was systematically lower, indicating extra series loss (for example inductor resistance and wiring parasitics) beyond the ideal model. The main shortcomings in this part of the experiment were treating the components as ideal, contact resistance and parasitics from the breadboard, and finite frequency resolution in the sweep. Future work using an LCR meter to characterise components over frequency, four–wire resistance measurements, a more rigid low–resistance circuit layout, and finer frequency steps near resonance would reduce these uncertainties and bring the measured Q closer to the theoretical prediction.

For the mechanical tuning fork, we again used ring–down and driven sweeps to extract f_0 and Q. The ring–down fit showed the fork to be an extremely weakly damped oscillator, with a well described decay and reduced chi–squared near one. In the driven measurements, the amplitude–ratio and phase fits were internally consistent with each other (similar f_0 in the tens of kHz and Q of order 18) and also showed small, structureless residuals, indicating that the same oscillator model describes that driven configuration. However, these driven values did not agree with the ring–down parameters: unlike the LCR case, we did not obtain a single (f_0, Q) from all three methods. A key experimental reason is that the driven data were taken on a different day from the ring–down, with the drive/pickup circuit and coil geometry rebuilt. It is likely that we did not reconstruct the mechanical–electrical setup exactly as intended, so the driven measurement effectively probed a different response of the system than the original ring–down. Additional shortcomings included the sensitivity of the fork–coil assembly to environmental vibrations and the strong dependence on coil alignment. In future experiments, using a rigid, vibrationisolated environment, fixing and documenting coil positions, and adopting a standardised procedure to verify circuit topology and probe polarity each time the setup is rebuilt would help ensure that ring–down and driven measurements probe the same mode and yield consistent parameters.

Altogether, these results show that an electronic LCR circuit and a mechanical tuning fork, while physically very different, are both well described by the same second–order differential equation for a damped, driven oscillator. Applying the same mathematical form to transient, amplitude, and phase data allowed us to extract resonant frequencies and quality factors in both systems, see where ideal models break down due to parasitic losses and setup errors, and illustrate how resonance and Q provide a common language for comparing electrical and mechanical oscillators.

References

[1] C. Lee, R. M. Serbanescu, and L. Helt, *Q Factor of Oscillators*, PHY324 Laboratory Manual, Department of Physics, University of Toronto, current revision March 17, 2025.

[2] Keysight Technologies, *InfiniiVision 1000 X-Series Oscilloscopes – Data Sheet*, 2018–2020.

[3] Wavetek Meterman, *Model 37XR Autoranging Digital Multimeter – Catalog Data Sheet*, RS Components stock no. 450–5278.

Appendix

Statement of AI Usage only autocomplete from Vscode no other AI usage

I Meterman 37XR Multimeter Uncertainty

For the Meterman 37XR multimeter, uncertainty depends on the measurement mode:

$$\sigma_{\text{reading}} = \begin{cases} 0.001 \times \text{reading} + 5 \times \text{sig} & \text{for voltage (V)} \\ 0.005 \times \text{reading} + 4 \times \text{sig} & \text{for resistance (R)} \\ 0.05 \times \text{reading} + 30 \times \text{sig} & \text{for inductance (I)} \\ 0.03 \times \text{reading} + 5 \times \text{sig} & \text{for capacitance (C)} \end{cases} \tag{17}$$

II Frequency Uncertainty

For frequency measurements using a Keysight 1000 X-Series oscilloscope:

$$\sigma_{f,\text{combined}} = \sqrt{\sigma^2_{f,\text{timebase}} + \sigma^2_{f,\text{display}}} \tag{18}$$

$$\text{where } \sigma_{f,\text{timebase}} = f \times \text{ppm} \times 10^{-6} \tag{19}$$

$$\sigma_{f,\text{display}} = 0.5 \times \text{LSB} \tag{20}$$

$$\sigma_{f,\text{relative}} = \frac{\sigma_{f,\text{combined}}}{f} \times 100\% \tag{21}$$

III Amplitude Ratio Uncertainty

For amplitude ratios $R = V_{\text{out}}/V_{\text{in}}$:

$$\sigma_R = R \times \sqrt{\left(\frac{\sigma_{V_{\text{out}}}}{V_{\text{out}}}\right)^2 + \left(\frac{\sigma_{V_{\text{in}}}}{V_{\text{in}}}\right)^2} \tag{22}$$

$$\tag{23}$$

IV Combined Uncertainty from Frequency and Amplitude

To propagate frequency uncertainty into amplitude uncertainty:

$$\sigma_{y,\text{total}} = \sqrt{\sigma^2_{y,\text{volt}} + \sigma^2_{y,\text{freq}}} \tag{24}$$

$$\text{where } \sigma_{y,\text{freq}} = \left|\frac{dM(f)}{df}\right| \times \sigma_f \tag{25}$$

$$\frac{dM(f)}{df} \approx \frac{M(f+\Delta f) - M(f-\Delta f)}{2\Delta f} \tag{26}$$